

Removing the subcritical-growth hypothesis from the high-energy multiplicity theorem in arXiv:2007.07024

Candidate full solution packet for `fa_banach_001`

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Status. Candidate full solution; likely valid, pending expert review.

1 Source problem

The source is V. Benci, S. Nardulli, L. E. Osorio Acevedo and P. Piccione, *Lusternik–Schnirelman and Morse Theory for the Van der Waals–Cahn–Hilliard equation with volume constraint*, arXiv:2007.07024, Nonlinear Analysis 220 (2022), Paper 112851 [1]. Its corrected photography argument appears in the subsequent corrigendum [2].

For a closed connected Riemannian manifold (M, g) , the constrained critical point equation is

$$-\varepsilon \Delta_g u + \frac{1}{\varepsilon} W'(u) = \lambda, \quad \int_M u \, dv_g = V. \quad (1)$$

The source proves the full multiplicity theorem under a subcritical upper growth condition on W' . If that upper-growth condition is replaced by

$$W''(s) \geq c_0 > 0 \quad (|s| \geq t_1 > 1), \quad (2)$$

Theorem 5.9 obtains only the low-energy counts $\text{cat}(M)$ and $P_1(M)$. Immediately before that theorem, on PDF page 28, the authors conjecture that the result remains true for high-energy solutions. The intended missing conclusions are those of their main theorem: $\text{cat}(M) + 1$, and $2P_1(M) - 1$ in the nondegenerate case.

2 Proof intuition

The source truncates a general potential to a subcritical one, but transfers only critical points whose energy is bounded: an energy estimate controls the Lagrange multiplier, after which the maximum principle keeps the solution in the unchanged part of the potential. The multiplier can instead be removed by comparing the global maximum and global minimum simultaneously.

Indeed, at a maximum and a minimum of a solution of (1),

$$W'(\max u) \leq \varepsilon \lambda \leq W'(\min u).$$

The fixed mean $a = V/\text{vol}_g(M)$ lies between these extrema. Since the derivative of the potential tends to $-\infty$ on the left and $+\infty$ on the right, one can place two fixed tail barriers $L < a < U$. A subcritical truncation is chosen to equal W on $[L, U]$, with derivative below every value attainable on $[a, \infty)$ to the left of L , and above every value attainable on $(-\infty, a]$ to the right of U . The displayed extrema inequality then forbids every truncated solution from leaving $[L, U]$, regardless of its energy. Thus the corrected subcritical multiplicity theorem transfers in its entirety.

we conclude that

$$|\lambda_{\varepsilon, V}| \leq \frac{2C\tilde{C}_0\rho^{-1}(1 + \varepsilon^{\frac{1}{2}}\rho^{-\frac{N}{2}})E_{\varepsilon}(u_{\varepsilon})}{\text{vol}_g(M) \left(1 - \left(\frac{V}{\text{vol}_g(M)}\right)^2\right)}. \quad \square$$

5.8. Remark. In Proposition 5.3 we do not require any subcritical growth condition, just that the second derivative of the potential is bounded from below by a positive constant in a neighborhood of infinity.

In our next result we generalize Theorem 2.1 by dropping assumption (2.3), and replacing it with a quite more general one. The price we pay is a weaker estimate on the lower bound on the number of solutions than the one determined in Theorem 2.1. In fact, our proof gives only the existence of low energy solutions. We conjecture that the following result remains true also for high energy solutions, but at the moment we are unable to give a complete proof. Thus under the assumptions of the preceding lemma we state the following theorem.

Figure 1: Original PDF page 28: the high-energy conjecture.

Theorem 5.9. *For every $W \in C^2(\mathbb{R})$ satisfying (2.2) and $W''(1) > 0$, $W''(s) > c_0 > 0$, $\forall |s| \geq t_1 > 1 > 0$, for some large t_1 , then for every $V \in]0, V^*[$ and $\varepsilon \in]0, \varepsilon^*[$, Problem (2.1) admits at least $\text{cat}(M)$ distinct solutions. Moreover, assume additionally that for given $V \in]0, V^*[$ and $\varepsilon \in]0, \varepsilon^*[$ all solutions of Problem (2.1) having energy less than or equal to the constant $\hat{c}$ defined in Lemma 5.1 are nondegenerate (see Definition 4.8). Then, Problem (2.1) has at least $P_1(M)$ distinct solutions.*

Proof. Let V^* be the constant determined in Theorem 2.1, and assume in the rest of the proof that $V \in]0, V^*[$. We can suppose that W satisfies

$$(5.19) \quad \limsup_{s \rightarrow +\infty} W'(s) = +\infty, \text{ and } \liminf_{s \rightarrow -\infty} W'(s) = -\infty,$$

because otherwise W' would be bounded, and so W would satisfy a growth condition as in (2.3), falling under the assumptions of Theorem 2.1. Now, using (5.19) it is easy to see

Figure 2: Original PDF page 29: the low-energy Theorem 5.9 that the conjecture seeks to upgrade.

3 Statement

We use the source's structural double-well assumptions (its condition (2.2) and the local barrier condition near the well at 1). Let $P_1(M) = \sum_k \beta_k(M)$, with coefficients in a fixed field.

Theorem 1 (High-energy extension without subcritical upper growth). *Let (M, g) be a closed connected Riemannian manifold and let $W \in C^2(\mathbb{R}, [0, \infty))$ satisfy the source's double-well assumptions and (2). Then there is $V^* > 0$ such that, for each $V \in (0, V^*)$, there is $\varepsilon^* > 0$ for which every $\varepsilon \in (0, \varepsilon^*)$ gives at least*

$$\text{cat}(M) + 1$$

distinct solutions of (1). If all solutions for the given (V, ε) are nondegenerate constrained critical points, then there are at least

$$2P_1(M) - 1$$

distinct solutions.

This restores exactly the low- plus high-energy conclusions of the corrected subcritical theorem under the general tail condition used in source Theorem 5.9.

4 Proof

We isolate the one-dimensional truncation and the energy-free range estimate.

Lemma 1 (A fixed admissible truncation). *Fix $a \in \mathbb{R}$. Under (2), there exist $L < a < U$ and a nonnegative C^2 double-well potential $\widehat{W}$ such that:*

1. $L < 0 < 1 < U$ and $\widehat{W} = W$ on $[L, U]$;
2. $\widehat{W}$ satisfies the subcritical upper-growth and tail-growth assumptions of the corrected main theorem in [1, 2];
3. if

$$A := \sup_{s \leq a} W'(s), \quad B := \inf_{s \geq a} W'(s),$$

then

$$\widehat{W}'(s) < B \quad (s < L), \quad \widehat{W}'(s) > A \quad (s > U).$$

Proof. Condition (2) implies $W'(s) \rightarrow -\infty$ as $s \rightarrow -\infty$ and $W'(s) \rightarrow +\infty$ as $s \rightarrow +\infty$. Hence $A < \infty$, $B > -\infty$, and $B \leq W'(a) \leq A$. Choose $L < \min\{0, a, -t_1\}$ and $U > \max\{1, a, t_1\}$ so that

$$W'(L) < B, \quad W'(U) > A.$$

On the left of L and the right of U , continue the endpoint two-jets by strictly convex C^2 tails. The left derivative is chosen increasing and strictly below B , while the right derivative is increasing and strictly above A . After a compact interpolation, the tails may be taken to have power growth $|s|^q$, where $2 < q < 2^* = 2 \dim(M)/(\dim(M) - 2)$ when $\dim(M) \geq 3$, and any finite $q > 2$ in dimensions one and two. The continuation grows away from the positive values $W(L)$ and $W(U)$, so it creates no new zero or critical well. This is the standard convex tail truncation used in Section 5 of [1], with the two strict derivative barriers retained. It has all the asserted properties. $\square$

Lemma 2 (All truncated critical points stay in the common core). *Put $a = V/\text{vol}_g(M)$ and choose $L, U, \widehat{W}$ from Lemma 1. Every solution (u, λ) of*

$$-\varepsilon \Delta_g u + \frac{1}{\varepsilon} \widehat{W}'(u) = \lambda, \quad \int_M u \, dv_g = V, \tag{3}$$

satisfies $L \leq u \leq U$ on M .

Proof. The subcritical growth of $\widehat{W}$ gives the standard elliptic regularity needed to use the maximum principle (alternatively, the same inequalities follow by testing with positive parts). Let

$$m = \min_M u, \quad R = \max_M u.$$

At a maximum, $\Delta_g u \leq 0$, whereas at a minimum $\Delta_g u \geq 0$. Equation (3) therefore yields

$$\widehat{W}'(R) \leq \varepsilon \lambda \leq \widehat{W}'(m). \tag{4}$$

The mean constraint gives $m \leq a \leq R$.

Suppose first that $R > U$. By Lemma 1, $\widehat{W}'(R) > A$. Since $m \leq a$, either $m \in [L, a]$, when $\widehat{W}'(m) = W'(m) \leq A$, or $m < L$, when $\widehat{W}'(m) < B \leq A$. In both cases $\widehat{W}'(R) > \widehat{W}'(m)$, contradicting (4). Thus $R \leq U$.

If $m < L$, then $\widehat{W}'(m) < B$. Since $R \geq a$, either $R \in [a, U]$, when $\widehat{W}'(R) = W'(R) \geq B$, or $R > U$, when $\widehat{W}'(R) > A \geq B$. Again $\widehat{W}'(R) > \widehat{W}'(m)$, contradicting (4). Thus $m \geq L$. $\square$

Proof of Theorem 1. Fix V in the small-volume range and set $a = V/\text{vol}_g(M)$. Let $\widehat{W}$ be the fixed subcritical potential supplied by Lemma 1. Apply the corrected subcritical multiplicity theorem [2, Theorem 2.1] to $\widehat{W}$. For sufficiently small ε , it produces at least $\text{cat}(M) + 1$ distinct solutions of (3).

Lemma 2 places the range of every one of these solutions in $[L, U]$, where $\widehat{W} = W$. Hence each pair also solves the original equation (1). This proves the first count. Notice that no energy bound has entered: the high-energy critical point transfers exactly as the low-energy ones do.

For the second count, assume all solutions of the original problem for the chosen (V, ε) are nondegenerate. Every solution of the truncated problem is an original solution by Lemma 2. Moreover $\widehat{W}''(u(x)) = W''(u(x))$ at every point, including a possible endpoint value because the extension is C^2 . Thus the constrained Hessians for the two functionals coincide at each truncated solution, so all truncated solutions are nondegenerate. The Morse part of the corrected subcritical theorem gives at least $2P_1(M) - 1$ of them, and all transfer to the original problem.

The small-volume constants in the corrected photography construction depend on the geometry and on the transition data on $[0, 1]$. These data are unchanged by the truncation; the fixed truncation may affect only the allowed ε^* , which is permitted to depend on V, M, g, W . This completes the proof. $\square$

5 Verification and limitations

The proof uses only two external inputs from the source: the corrected subcritical multiplicity theorem and the standard existence of a subcritical convex tail truncation. The new step is Lemma 2. Its signs have been checked directly at both extrema, and both possible locations of the opposite extremum (inside or outside the common core) are included explicitly. No numerical computation is involved.

The conclusion is limited to the source's tail-convex class (2); it does not assert a result for a potential whose derivative fails to tend to the corresponding infinities. It also inherits the small-volume and small- ε regime of the corrected photography theorem.

6 Novelty check

Before promotion, the four cheap run indexes were searched for the arXiv id, title, and the core high-energy/subcritical-growth terms. Bounded external searches through 11 August 2026 used the exact conjecture sentence, the title, the terms “high energy solutions”, “dropping the subcritical growth condition”, and the extrema inequality. They found the original article and its corrigendum, but no later solution. The 2026 survey by Andrade, Nardulli and Ponciano [3] still presents the closed scalar theorem for a subcritical potential and does not record this extension. This is evidence, not an exhaustive bibliographic guarantee.

7 Human-review recommendation

Review Lemma 2 first; it is the decisive and elementary step. Then check that the one-dimensional continuation in Lemma 1 is chosen within the exact admissible class used by the corrected theorem. If both checks pass, the packet gives a full resolution of the Section 5 high-energy conjecture, including its Morse-theoretic count.

References

- [1] V. Benci, S. Nardulli, L. E. Osorio Acevedo and P. Piccione, *Lusternik–Schnirelman and Morse Theory for the Van der Waals–Cahn–Hilliard equation with volume constraint*, Nonlinear Analysis 220 (2022), Paper 112851; arXiv:2007.07024.
- [2] V. Benci, D. Corona, S. Nardulli, L. E. Osorio Acevedo and P. Piccione, *Corrigendum to: “Lusternik–Schnirelman and Morse Theory for the Van der Waals–Cahn–Hilliard equation with volume constraint”*, Nonlinear Analysis 238 (2024), Paper 113389; latest arXiv:2007.07024.
- [3] J. H. Andrade, S. Nardulli and R. Ponciano, *Survey on topological methods for Allen–Cahn equations and systems*, arXiv:2604.23920 (2026).